



FD-X Series Setting guide

(For precise detection of shot amounts)



Overview of the settings

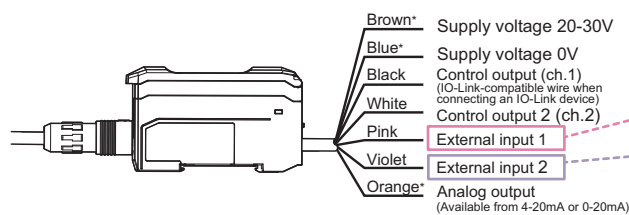
The following describes recommended settings for precise detection of shot amounts.
For details of each setting, refer to the instruction manual included with the product.

Setting items	Descriptions	Reference pages
(1) Initial settings	These settings are selected when power is turned ON for the first time.	2
(2) Origin adjustment	Once attached to the pipe, this is used to adjust the reading for zero flow.	2
(3) Configuring functions - Detection mode of the ch. 1 : Shot mode - Selecting input 1 function : Shot sample - Selecting input 2 function : Zero shift - Response time : 50ms - Display resolution (discharge rate) : lower it if necessary - Zero cut flow rate : Near 0mL/min * - Shot decrement : On	Apply shot mode, which determines a threshold and OK/NG judgment for each shot. Select zero shift for an external input so that continuous changes of the origin can be corrected. * It is recommended to set this value to something low but not necessarily zero, as 3 mL / min, because slight variation could lead to reading errors.	3 to 5
(4) Target calibration	The span is adjusted based on the shot amount.	6

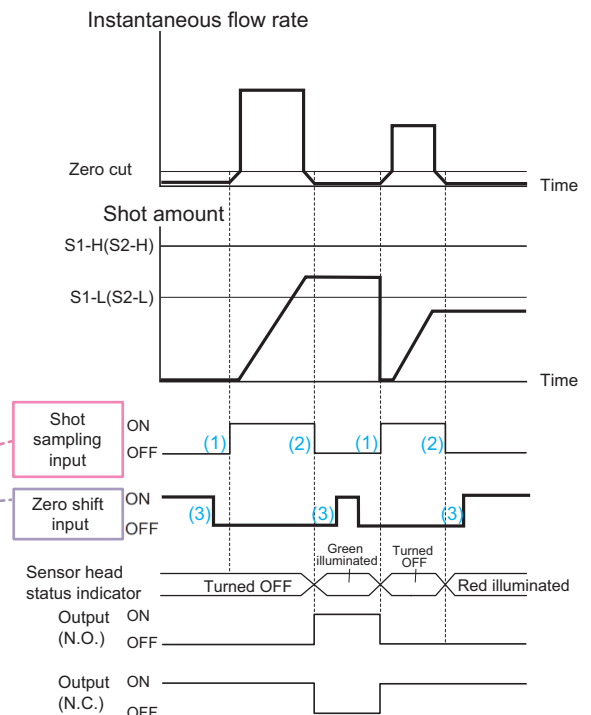
Description of the operation

The shot amount is detected while the shot sampling input (input 1) is ON and OK/NG is determined.

- The shot amount is reset when the shot sampling input is turned ON and the detection starts.
- Detection stops when the shot sampling input turns OFF and OK/NG is determined according to the set thresholds.
- More accurate detection is possible by turning ON the zero shift input and carrying out the origin adjustment while shots are not occurring.



* There is no brown, blue, or orange wires with the FD-XA2 (expansion unit).

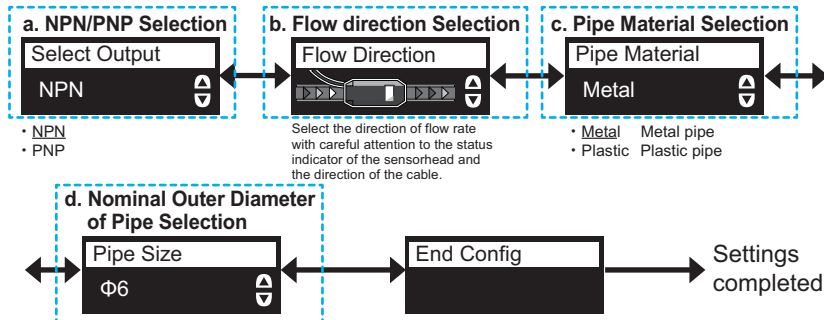


(1) Initial settings

Press or to move to the next menu screen.

Press to return to the previous menu screen.

Press or to select a parameter.



Select the appropriate parameters according to the flow direction and the pipe type.

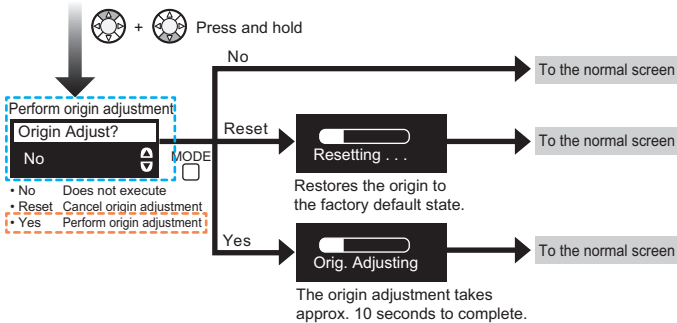
FD-XS1	FD-XS8	FD-XS20
φ 3	φ 6	φ 10
φ 4	φ 8	φ 12
1/8"(3.18mm)	1/4"(6.35mm)	3/8"(9.53mm)
		1/2"(12.7mm)
		6A(10.5mm)
		8A(13.8mm)

(2) Origin adjustment (press and hold the UP and DOWN buttons)

This function sets the instantaneous flow rate at the time of performing the origin adjustment as "zero". Perform the origin adjustment after completing the installation and initial settings.

Point Perform this function when the pipe is filled with fluid and the fluid is not moving.

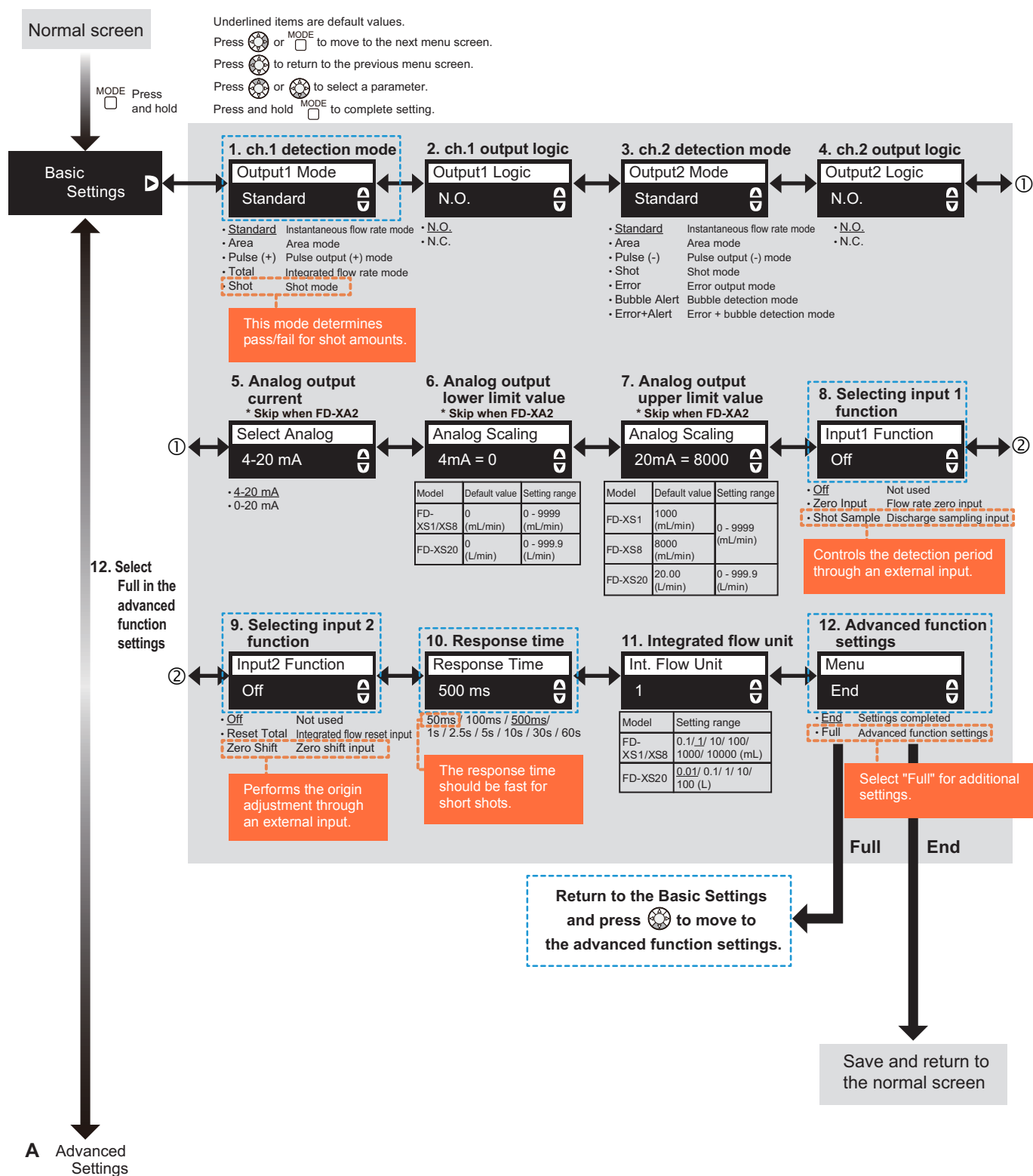
Normal screen (Instantaneous flow rate)



The origin adjustment takes about 10 seconds. The result will be stored after the power is turned off.
(Adjustment with the zero shift of an external input will be reset when the power is turned off.)

Reference If the FD-X series fails to adjust the origin, it shows "Error" and goes back to the normal screen.

(3) Configuring functions (press and hold the MODE button)



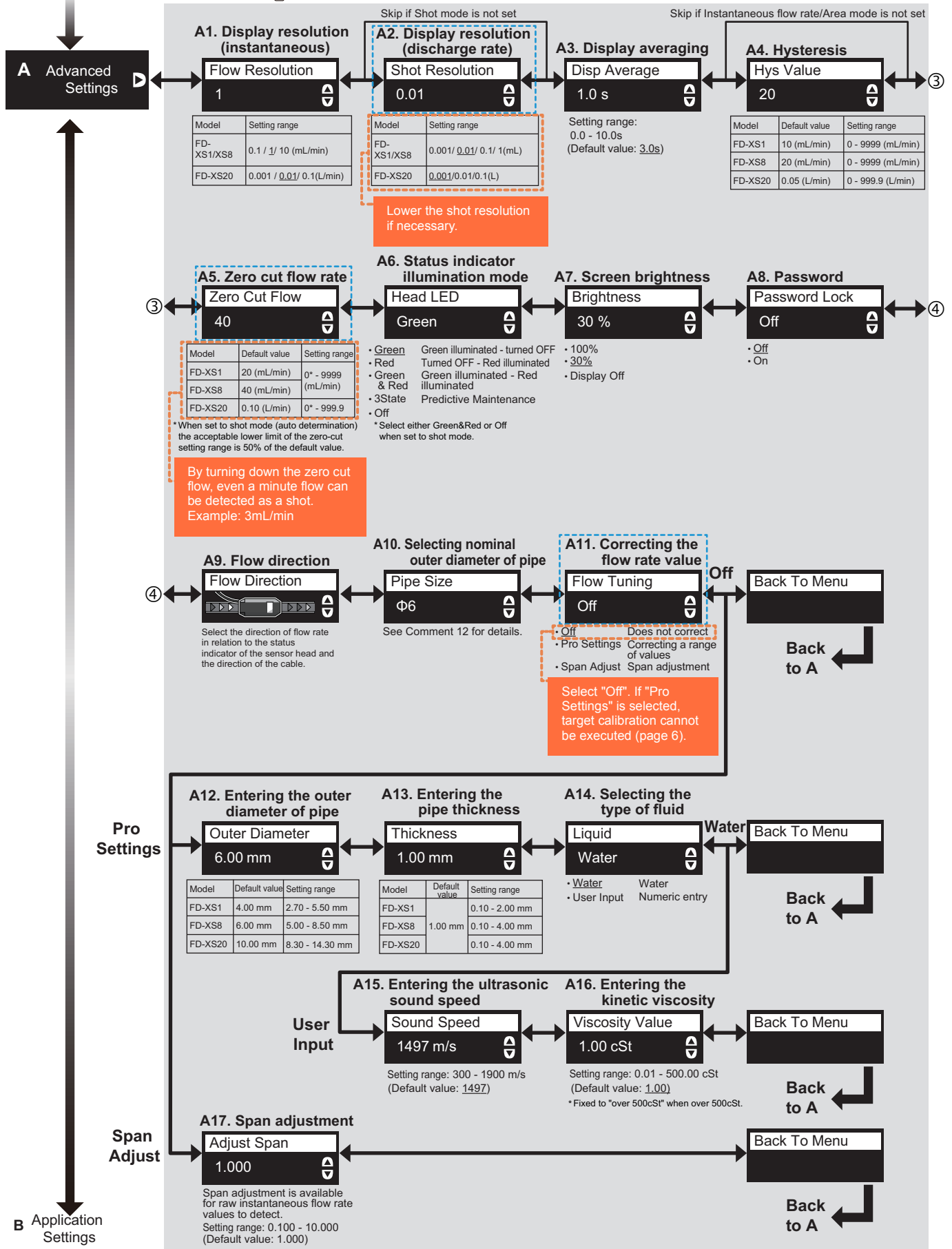
Underlined items are default values.

Press  or MODE to move to the next menu screen.

Press  to return to the previous menu screen.

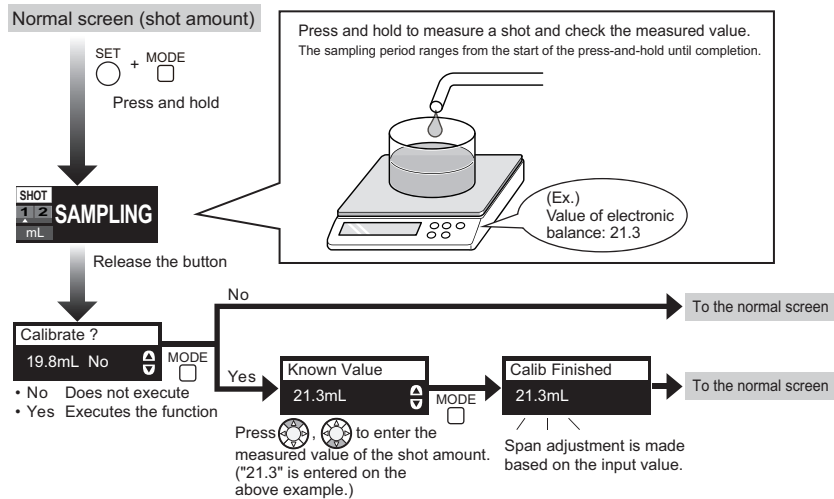
Press  or  to select a parameter.

Press and hold MODE to complete setting.



(4) Target calibration (press and hold the SET and MODE buttons with the shot amount screen displayed)

This feature is used to correct the flow rate by comparing the reading value to a known shot amount.



Reference If the FD-X series fails to calibrate the target, it shows "Error" and goes back to the normal screen.

By executing the target calibration, the span is adjusted automatically.
The graph on the right is an image of the span adjustment.
You can check and change the current span value with "A17. Span adjustment" (page 4).

